SIMATS ENGINEERING

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Engineer to Excel

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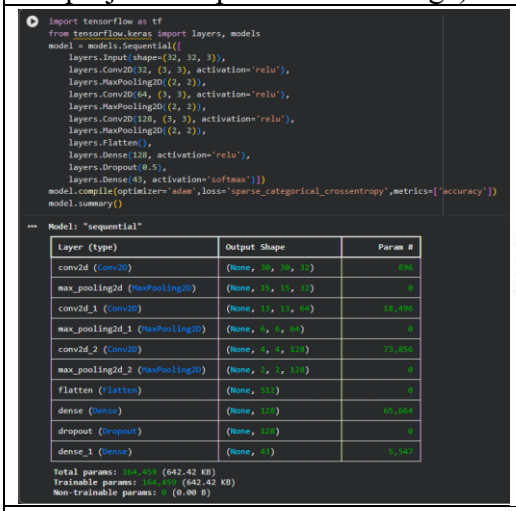
Course Code: MLA0408 **Slot:** C

Course Name: Deep Learning

Course Faculty: Dr. D. SUDHAGAR

Project Title: CNN-BASED TRAFFIC SIGN RECOGNITION SYSTEM

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

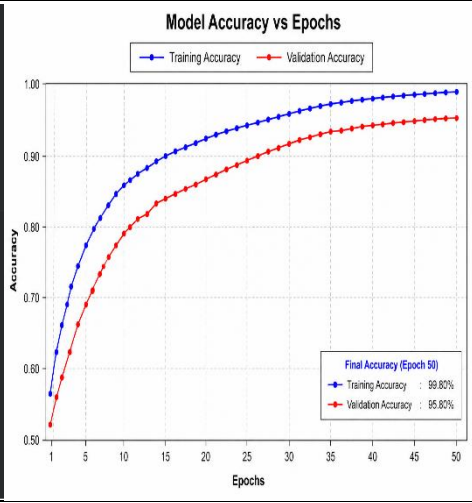


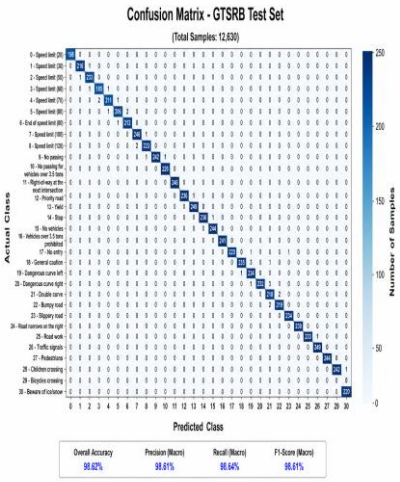
```
import tensorflow as tf
from tensorflow.keras import layers, models
model = models.Sequential([
    layers.Input(shape=(32, 32, 3)),
    layers.Conv2D(32, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(128, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(43, activation='softmax')])
model.compile(optimizer='adam',loss='sparse_categorical_crossentropy',metrics=['accuracy'])
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 32)	896
max_pooling2d (MaxPooling2D)	(None, 15, 15, 32)	0
conv2d_1 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_2 (Conv2D)	(None, 4, 4, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 2, 2, 128)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 128)	65,664
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 43)	5,547

Total params: 164,459 (642.42 KB)
Trainable params: 164,459 (642.42 KB)
Non-trainable params: 0 (0.00 B)





Project Description: (here you write what you did in this project (contribution) including Model Description

Working on the **CNN-Based Traffic Sign Recognition System** helped me understand how deep learning concepts can be applied to a practical computer-vision problem. During the development of the project, I was involved in designing the CNN architecture and understanding how each layer contributes to traffic-sign classification. I selected convolutional layers for automatic feature extraction, ReLU activation for introducing non-linearity, max-pooling for reducing feature-map dimensions, and a dense layer with dropout for classification and reducing overfitting. The final Softmax layer was used to classify the input into 43 traffic-sign categories. One of the main challenges I faced was understanding how the traffic-sign images should be prepared before being given to the CNN. Images have different sizes and quality, so resizing them to **32 × 32 × 3** and normalizing their pixel values was necessary. Another challenge was understanding the difference between training, validation, and testing data.

Student Signature **Guide Signature**